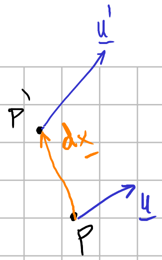


$$\left.\begin{array}{l} \underline{u} = \underline{u}(P) \\ \underline{u}' = \underline{u}(P') \end{array}\right\} \quad \underline{u}' - \underline{u} = d\underline{u} = \nabla \underline{u} \cdot d\underline{x}$$

$$(\underline{u}' - \underline{u})_i = (d\underline{u})_i = (\nabla \underline{u} \cdot d\underline{x})_i = \frac{\partial u_i}{\partial x_j} \cdot dx_j = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_i}{\partial x_j}\right) dx_j =$$

$$= \left[\frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \underbrace{\frac{\partial u_j}{\partial x_i} - \frac{\partial u_j}{\partial x_i}}_{0} + \frac{\partial u_i}{\partial x_j}\right)\right] dx_j =$$

$$= \left[\underbrace{\frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)}_{E_{ij}} + \underbrace{\frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right)}_{-\omega_{ij}}\right] dx_j =$$

$\underline{\underline{E}} = \underline{\underline{E}}^T$

$E_{ij} = E_{ji}$

$E_{11} = E_{11}$

$E_{12} = E_{21}$

$$\begin{bmatrix} E_{11} & E_{12} \\ E_{12} & E_{22} \end{bmatrix}$$

$$= (E_{ij} + \omega_{ji})\, dx_j$$

Si fuera velocidad $(\underline{v}' - \underline{v})_i = (d\underline{v})_i = (V_{ij} + \Omega_{ji})\, dx_j$

$-\underline{\underline{\omega}} = \underline{\underline{\omega}}^T$

$-\omega_{ij} = \omega_{ji}$

$-\omega_{11} = \omega_{11} = 0$

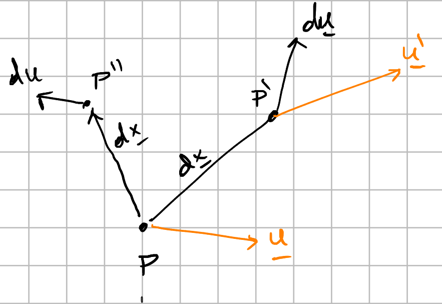
$-\omega_{12} = \omega_{21}$

$$\begin{bmatrix} 0 & \omega_{12} \\ -\omega_{12} & 0 \end{bmatrix}$$

$$\underline{u}' - \underline{u} = d\underline{u} = \nabla \underline{u} \cdot d\underline{x} = (\underline{\underline{E}} - \underline{\underline{\omega}})\, d\underline{x}$$

Evaluar el movimiento del cuerpo como cuerpo rígido: $\Rightarrow \underline{\underline{E}} = 0$

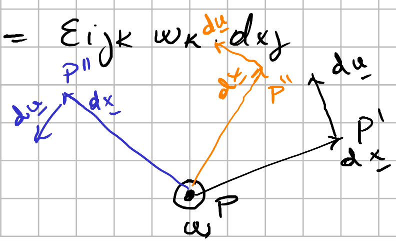
$$\underline{u}' - \underline{u} = d\underline{u} = -\underline{\underline{\omega}}\, d\underline{x}$$



$$\omega_{ij} = \varepsilon_{ijk}\, \omega_k$$

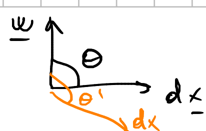
$$(\underline{u}' - \underline{u})_i = u_i' - u_i = du_i = -\omega_{ij}\, dx_j = \varepsilon_{ijk}\, \omega_k\, dx_j$$

$$d\underline{u} = \underline{\omega} \times d\underline{x}$$



$$|d\underline{u}| = |\underline{\omega}|\, |d\underline{x}| \cdot \sin(\theta)$$

si asumimos $\theta = \pi/2$



b)

$$\omega_k = \frac{1}{2}\varepsilon_{kij}\omega_{ij} \quad \rightarrow \quad \omega_{ij} = \varepsilon_{ijk}\omega_k$$

$$\delta_{pi}\,\omega_{ij}$$

$$\omega_k = \frac{1}{2}\varepsilon_{kij}\omega_{ij} \qquad \times\ \varepsilon_{pqk}$$

$$\underbrace{\delta_{p1}\omega_{1j}}_{\delta_{11}\omega_{1j}} + \underbrace{\delta_{p2}\omega_{2j} + \delta_{p3}\omega_{3j}}_{\cancel{\delta_{21}}^{0}\omega_{1j} + \cancel{\delta_{31}}^{0}\omega_{1j} + \dots}$$

$$\varepsilon_{pqk}\,\omega_k = \frac{1}{2}\varepsilon_{pqk}\varepsilon_{kij}\omega_{ij} =$$

$$\delta_{21} = 0$$

$$\delta_{11} = 1$$

$$= 1/2\ \varepsilon_{kpq}\varepsilon_{kij}\omega_{ij}$$

$$= \frac{1}{2}\left(\delta_{pi}.\delta_{qj} - \delta_{pj}\delta_{qi}\right)\omega_{ij}$$

$$= \frac{1}{2}\left(\omega_{pq} - \omega_{qp}\right) =$$

$$= \frac{1}{2}\left(\omega_{pq} + \omega_{pq}\right) = \omega_{pq}$$

$$\omega_{pq} = \varepsilon_{pqk}\,\omega_k$$

d)

$$\underline{\Omega} = \operatorname{curl} \underline{v} = \underline{\nabla} \times \underline{v}$$

$$\Omega_{ij} = \frac{1}{2}\left(\frac{\partial v_j}{\partial x_i} - \frac{\partial v_i}{\partial x_j}\right)$$

$$\Omega_K = \varepsilon_{Kij} \cdot \Omega_{ij} = \varepsilon_{Kij} \cdot \frac{1}{2}\left(\frac{\partial v_j}{\partial x_i} - \frac{\partial v_i}{\partial x_j}\right) =$$

$$= \frac{1}{2}\left(\varepsilon_{Kij} \cdot \frac{\partial v_j}{\partial x_i} - \varepsilon_{Kij} \cdot \frac{\partial v_i}{\partial x_j}\right) = \frac{1}{2}\left(\boxed{\varepsilon_{Kij} \cdot \frac{\partial v_j}{\partial x_i}} + \boxed{\varepsilon_{Kji} \cdot \frac{\partial v_i}{\partial x_j}}\right)$$

$$= \varepsilon_{Kij} \cdot \frac{\partial v_j}{\partial x_i} = (\underline{\nabla} \times \underline{v})_K$$

$$\underline{\Omega} = \underline{\nabla} \times \underline{v}$$